Lecture 11

Taylor approximation and small oscillations

Uses.

Summary 2: Finite polynomial algebra and the triangle inequality control sums.
Summary 4: The constructed number e satisfies $2 < e < 3$.
Summary 5: Continuous compositions preserve limits. A function with a finite limit is bounded near the approach point. Positive powers have the proved continuity properties on their stated domains.
Summary 7: Differentiability gives a linear approximation with remainder $o(h)$, meaning a remainder whose quotient by h tends to zero.
Summary 8: The derivative rules and elementary table are proved.
Summary 9: Higher derivatives and the classes C^n are defined.
Summary 10: Cauchy's mean value theorem compares two function changes. A positive second derivative at a stationary point gives a strict local minimum.

Objectives. *A Taylor polynomial matches several derivatives at one point. The remainder formula turns that matching into an error estimate. We prove the formula, compare the meanings of bounded and negligible remainders, and examine a function with derivatives of every order whose Taylor coefficients all vanish. A quadratic potential approximation then gives a linearised mechanical equation with a stated Newton-law premise.*

By the end of this lecture you will be able to:

- Construct Taylor polynomials and prove the Lagrange remainder for C^{n+1} functions.
- State error orders with O and o and prove the Peano remainder for C^n functions.
- Bound approximation errors and explain the zero-coefficient example with derivatives of every order.
- Linearise the stated Newton model with positive mass at a C^2 nondegenerate potential minimum.

11.1 Matching derivatives at a point

For integers $n \geq 1$, the *factorial* is $n! = 1 \cdot 2 \cdots n$. We set $0! = 1$. The notation $\sum_{k=0}^{n}$ adds the terms with indices $0, 1, \ldots, n$ and is a finite sum. We also write $C^0(I)$ for the continuous functions on an interval I. For positive orders, $C^n(I)$ has the meaning given in Lecture 9.

Let a be an interior point of a real interval and suppose f has derivatives through order n

at a, where $n \geq 0$. The *Taylor polynomial of order n at a* is

$$T_{n,a}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k. \tag{11.1}$$

Its degree is at most n, since a highest coefficient can be zero. As before, a zeroth power in a polynomial is the constant 1, including at its centre. We use $f^{(0)} = f$.

This polynomial is the unique polynomial of degree at most n whose derivatives through order n, including the function value, agree with those of f at a. Indeed, any such polynomial can be written as $Q(x) = \sum_{k=0}^{n} c_k(x-a)^k$ by substituting $x = (x-a)+a$ in its ordinary polynomial expression. Repeated differentiation and evaluation at a give $Q^{(k)}(a) = k!c_k$. Thus matching forces $c_k = f^{(k)}(a)/k!$, exactly the displayed coefficients. Conversely, these coefficients give the required derivative values.

For example, every derivative of e^x is e^x, so at zero its Taylor polynomial of order 3 is $1 + x + x^2/2 + x^3/6$. For sine, the derivative values through order 3 at zero are $0, 1, 0, -1$. Its corresponding polynomial is $x - x^3/6$. Matching derivatives alone has not yet bounded either approximation error.

11.2 The Lagrange remainder

Write the *remainder* as a function of the increment:

$$R_{n,a}(h) = f(a+h) - T_{n,a}(a+h).$$

The next theorem represents this actual difference at a nonzero increment.

Theorem 11.1 (Taylor's formula with Lagrange remainder)**.** *Let $n \geq 0$ be an integer and let $f \in C^{n+1}(I)$ on an open real interval I. For $a, a+h \in I$ and $h \neq 0$, some point ξ strictly between a and $a+h$ satisfies*

$$f(a+h) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}h^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}h^{n+1}.$$

If $|f^{(n+1)}| \leq M$ between the two inputs, then

$$|R_{n,a}(h)| \leq \frac{M}{(n+1)!}|h|^{n+1}.$$

The error bound also holds at $h = 0$, where the remainder is zero.

Proof. Fix $b = a + h$ and define, for t between a and b,

$$G(t) = f(b) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(b-t)^k, \qquad H(t) = (b-t)^{n+1}.$$

Both functions are continuous on the closed interval with those endpoints and differentiable inside. Their endpoint values are $G(b) = H(b) = 0$, $G(a) = R_{n,a}(h)$ and $H(a) = h^{n+1}$.

The derivative of the sum in G is

$$\sum_{k=0}^{n} \frac{f^{(k+1)}(t)}{k!}(b-t)^k - \sum_{k=1}^{n} \frac{f^{(k)}(t)}{(k-1)!}(b-t)^{k-1}.$$

In the second sum, replacing the index by $k-1$ gives the same terms as the first sum except its final term. They cancel, so

$$G'(t) = -\frac{f^{(n+1)}(t)}{n!}(b-t)^n, \qquad H'(t) = -(n+1)(b-t)^n.$$

For $n = 0$, the second sum is empty and the same formulas hold directly. The derivative H' is nonzero in the open interval, since $t \neq b$. Cauchy's theorem, Theorem 10.1, therefore supplies an interior ξ with

$$\frac{R_{n,a}(h)}{h^{n+1}} = \frac{G(b)-G(a)}{H(b)-H(a)} = \frac{G'(\xi)}{H'(\xi)} = \frac{f^{(n+1)}(\xi)}{(n+1)!}.$$

For $h < 0$, use the reversed endpoint order in Cauchy's theorem. Both differences change sign, so their quotient is unchanged. Multiplying proves the remainder formula. Its absolute value and the assumed derivative bound give the error estimate. ∎

For $f(x) = e^x$, order 3 at zero and $h = 1/10$ give

$$T_{3,0}(1/10) = \frac{6631}{6000}, \qquad 0 < R_{3,0}(1/10) < \frac{3}{24 \cdot 10^4} = \frac{1}{80000}.$$

Here the fourth derivative is e^x and the intermediate input lies between 0 and $1/10$, so its value is positive and below $e < 3$. Thus the computed rational number is a lower approximation, with the explicit upper error $1/80000$.

11.3 Bounded and negligible error orders

Let r and s be real functions near zero, with $s(h) \neq 0$ for all sufficiently small nonzero h. The notation $r(h) = o(s(h))$ means $r(h)/s(h) \to 0$. This extends Lecture 7's $o(h)$ notation. The notation $r(h) = O(s(h))$ means that some constants $C > 0$ and $\delta > 0$ satisfy

$$|r(h)| \leq C|s(h)| \qquad (0 < |h| < \delta).$$

Little-oh says that the relative error tends to zero. Big-oh says that the relative error is bounded. Little-oh implies big-oh by taking a radius where the quotient has absolute value below 1. The converse fails: $h^2 = O(h^2)$, but h^2 is not $o(h^2)$ because its quotient by itself is 1.

The Lagrange bound gives $R_{n,a}(h) = O(h^{n+1})$ locally when $f \in C^{n+1}$ near a. Indeed, the continuous derivative of order $n+1$ is bounded on a sufficiently small closed interval around a by the extreme value theorem. One bound M then works for every sufficiently small increment. A negligible remainder of order n needs only C^n regularity.

Theorem 11.2 (Peano remainder). *If $f \in C^n(I)$ with $a \in I$, where I is open and $n \geq 0$ is an integer, then*

$$f(a+h) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}h^k + o(h^n) \qquad (h \to 0).$$

Proof. For $n = 0$, continuity gives $f(a+h) - f(a) \to 0$, which is $o(1)$. For $n \geq 1$, apply the proved Lagrange formula at order $n-1$. The required regularity is exactly C^n. Subtracting the additional order-n Taylor term gives

$$R_{n,a}(h) = \frac{f^{(n)}(\xi) - f^{(n)}(a)}{n!}h^n,$$

where ξ lies between a and $a+h$. Thus $|\xi - a| \leq |h|$. Continuity of $f^{(n)}$ at a makes the coefficient tend to zero, proving $R_{n,a}(h)/h^n \to 0$. ∎

11.4 Elementary expansions and sine error bounds

The sine and cosine derivatives repeat after four differentiations. The exponential is unchanged by every differentiation. For a fixed $\alpha \in \mathbb{R}$ and every integer $k \geq 1$, repeated differentiation gives

$$\frac{d^k}{dx^k}\ln(1+x) = \frac{(-1)^{k-1}(k-1)!}{(1+x)^k}, \qquad \frac{d^k}{dx^k}(1+x)^\alpha = \alpha(\alpha-1)\cdots(\alpha-k+1)(1+x)^{\alpha-k}.$$

The logarithm formula follows by induction from its first derivative. Differentiating its right side multiplies by $-k$ and raises the denominator power by one, giving the next factorial and sign. The real-power formula follows in the same way: the next differentiation adds the factor $\alpha - k$ and lowers the power by one. Both are used on $x > -1$, so every power base is positive.

Write $\binom{\alpha}{0} = 1$ and, for integers $k \geq 1$, define

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}.$$

These are the *generalised binomial coefficients*. Substituting the derivative values at zero gives the table below. The remainders are as $x \to 0$. Each needed derivative is continuous near zero, so the Lagrange bound applies. Sine's order-4 coefficient also vanishes, giving its indicated order-5 error.

Function	Taylor polynomial shown	Remainder
e^x	$1 + x + x^2/2 + x^3/6$	$O(x^4)$
$\sin x$	$x - x^3/6$	$O(x^5)$
$\cos x$	$1 - x^2/2$	$O(x^4)$
$\ln(1+x)$	$x - x^2/2 + x^3/3$	$O(x^4)$
$(1+x)^\alpha$	$1 + \alpha x + \binom{\alpha}{2}x^2 + \binom{\alpha}{3}x^3$	$O(x^4)$

For a quantitative sine estimate, use Taylor order 2 at zero. Its polynomial is x, since the value and second derivative of sine at zero are both zero. The third derivative has absolute value at most 1, so

$$|\sin x - x| \leq \frac{|x|^3}{6} \qquad (x \in \mathbb{R}). \tag{11.2}$$

Order 4 gives the same polynomial as order 3 and the sharper cubic estimate

$$\left|\sin x - \left(x - \frac{x^3}{6}\right)\right| \leq \frac{|x|^5}{120} \qquad (x \in \mathbb{R}).$$

Figure 11.1 compares these approximations.

For a nonzero radian angle with $|\theta| \leq 1/10$, Eq. (11.2) gives

$$\frac{|\sin\theta - \theta|}{|\theta|} \leq \frac{\theta^2}{6} \leq \frac{1}{600}.$$

At zero both expressions are zero. The relative quotient is used only at nonzero angles.

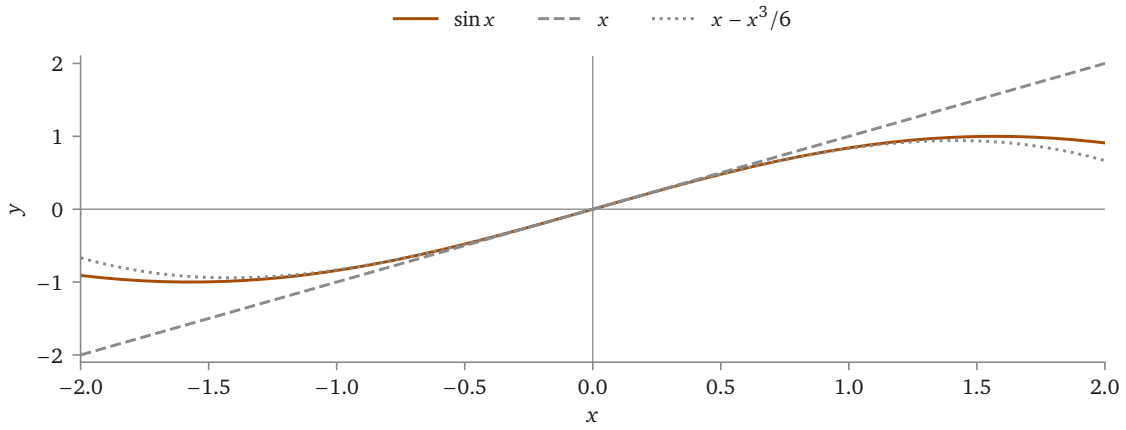


Figure 11.1: Sine and its linear and cubic Taylor polynomials at zero. The remainder estimates quantify their errors away from the centre.

11.5 A function with zero Taylor coefficients

A function is *infinitely differentiable* if it has continuous derivatives of every nonnegative integer order. The class of these functions on an interval I is denoted $C^\infty(I)$. Define $F : \mathbb{R} \to \mathbb{R}$ by $F(0) = 0$ and $F(x) = e^{-1/x^2}$ for $x \neq 0$. We prove that F is C^∞ and $F^{(j)}(0) = 0$ for every integer $j \geq 0$.

First, Taylor's formula for e^u at zero has positive remainder for $u > 0$. All polynomial terms are nonnegative. For each integer $N \geq 0$,

$$e^u \geq \frac{u^N}{N!} \quad (u > 0), \qquad 0 < e^{-1/h^2} \leq N!|h|^{2N} \quad (h \neq 0).$$

For nonzero x, repeated differentiation gives $F^{(j)}(x) = P_j(1/x)e^{-1/x^2}$ with a polynomial P_j. The construction starts at $P_0 = 1$ and satisfies

$$P_{j+1}(u) = -u^2 P_j'(u) + 2u^3 P_j(u).$$

For example, $P_1(u) = 2u^3$ gives $F'(x) = 2x^{-3}e^{-1/x^2}$ for $x \neq 0$. Indeed, differentiating $u = 1/x$ gives $-u^2$, while the exponential's derivative gives the factor $2u^3$. Induction shows that P_j uses only powers from 0 through $3j$: the two terms in the recurrence have powers at most $3j + 1$ and $3j + 3$. The triangle inequality therefore gives a constant $C_j > 0$ such that $|P_j(1/h)| \leq C_j|h|^{-3j}$ when $0 < |h| \leq 1$.

Now argue by induction at zero. The base value is $F(0) = 0$. Suppose the jth derivative exists there with value zero. Taking $N = 2j + 1$ in the exponential bound gives

$$\left|\frac{F^{(j)}(h) - F^{(j)}(0)}{h}\right| \leq C_j(2j + 1)!|h|^{j+1} \longrightarrow 0.$$

Thus $F^{(j+1)}(0)$ exists and equals zero. Away from zero its derivative has the next polynomial form already computed. Induction gives derivatives of every order everywhere. Each derivative is continuous because it is itself differentiable. Hence F is C^∞.

Every Taylor polynomial of F at zero is the zero polynomial, although $F(x) > 0$ for every nonzero x. Thus functions with derivatives of every order can have identical Taylor polynomials at a point and different values throughout its deleted neighbourhood. The earlier bound also gives $F(h) = o(h^k)$ for every fixed integer $k \geq 0$, by choosing $2N > k$ and dividing by $|h|^k$.

11.6 A quadratic potential and a linearised oscillator

Let $I, J \subseteq \mathbb{R}$ be open intervals, and let $q : J \to I$ be a twice differentiable position as a function of time. Let $m > 0$ be a constant mass and let $V : I \to \mathbb{R}$ be C^2. The function V is a *potential energy* for the force $-V'(q)$. Dots denote derivatives with respect to time. We assume the physical premise of Newton's equation

$$m\ddot{q} = -V'(q).$$

This is a model assumption. The conclusions below are deductions within it. Suppose $q_0 \in I$ satisfies

$$V'(q_0) = 0, \qquad V''(q_0) = k > 0.$$

The constant position $q = q_0$ solves Newton's equation and is an *equilibrium*. The second-derivative test makes q_0 a strict local minimum of V. With positive second derivative, this is called a *nondegenerate minimum*.

Put $h = q - q_0$. The proved C^2 Peano formula gives

$$V(q_0 + h) = V(q_0) + \frac{k}{2}h^2 + o(h^2).$$

Apply the linear approximation criterion directly to V':

$$V'(q_0 + h) = kh + \eta(h), \qquad \frac{\eta(h)}{h} \longrightarrow 0.$$

For sufficiently small nonzero h, the quotient $\eta(h)/h$ has absolute value below $k/2$. Thus $V'(q_0 + h)$ has the sign of h, and the force points towards the equilibrium. Newton's exact equation becomes $m\ddot{h} + kh = -\eta(h)$. Keeping the linear force gives the *linearised equation*

$$m\ddot{h} + kh = 0.$$

Set

$$\omega_0 = \sqrt{k/m}. \tag{11.3}$$

For any constants $A, B \in \mathbb{R}$, substitution of $h(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$ gives $\ddot{h} = -\omega_0^2 h$, hence the linearised equation. For $(A, B) \neq (0, 0)$, the positive constant ω_0 is the *angular frequency* of these sinusoidal motions. The pair $(A, B) = (0, 0)$ gives the equilibrium. Their terms repeat after time

$$T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}. \tag{11.4}$$

These formulas concern the linearised equation. An error bound between its trajectories and nonlinear trajectories over a time interval would require a separate comparison argument. Lecture 34 proves the general solution statement for the linear equation.

For example, in chosen units take $m = 2$ and $V(q) = q^2/2 + q^4/4$. Then $q_0 = 0$, $V'(q) = q + q^3$ and $k = V''(0) = 1$. The linearised equation is $2\ddot{h} + h = 0$, with $\omega_0 = 1/\sqrt{2}$ and $T_0 = 2\pi\sqrt{2}$. The exact force differs from $-h$ by $-h^3$. For nonzero displacements its error relative to that linear force has magnitude h^2, which tends to zero. Figure 11.2 compares the potential with its quadratic part.

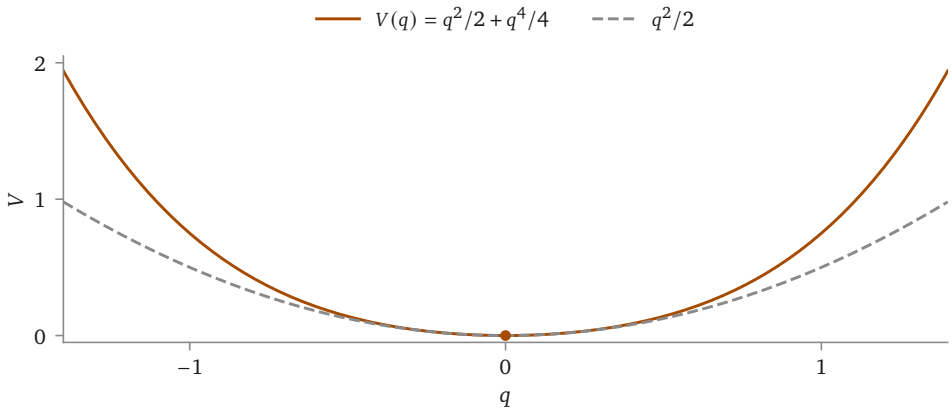


Figure 11.2: The potential $q^2/2 + q^4/4$ and its quadratic part $q^2/2$. They have the same value, first derivative and positive curvature at zero.

Summary 11.

- Equation (11.1): the Taylor polynomial of order n is the unique polynomial of degree at most n matching derivatives through order n at the centre.
- Theorem 11.1: a C^{n+1} function has Lagrange remainder $f^{(n+1)}(\xi)h^{n+1}/(n+1)!$, with ξ between the two inputs. A bound on that derivative gives a quantitative error bound.
- Theorem 11.2: C^n regularity gives remainder $o(h^n)$. Big-oh expresses a bounded relative error, while little-oh makes that error vanish.
- The function e^{-1/x^2}, extended by zero, is C^∞ with every Taylor coefficient at zero equal to zero. It is positive off zero.
- Equation (11.3): the stated Newton model near a C^2 nondegenerate potential minimum has linearised angular frequency $\sqrt{V''(q_0)/m}$. The derivation retains both the physical premise and $m > 0$.

Sheet 11 applies expansions to a composition and to relativistic energy, proves irrationality of e from a finite remainder, and tests a degenerate minimum. Lecture 12 introduces vectors and their lengths and projections.

References

V. A. Zorich, *Mathematical Analysis I*, 2nd ed., Springer, 2015, §5.3. S. R. Ghorpade and B. V. Limaye, *A Course in Calculus and Real Analysis*, 2nd ed., Springer, 2018, §4.4.